

Scalability & Future Plan Template

Date	15 March 2026
Team ID	Batch 2023-27 ACOE
Project Name	Rising Waters – ML-Based Flood Prediction System
Maximum Marks	1 Mark

Current Scalability Architecture

Aspect	Current Implementation	Scale-Up Path
Hosting	IBM Cloud Foundry (single instance)	Scale to multiple CF instances
Container	Docker (Dockerfile provided)	Kubernetes orchestration via IBM CKS
Storage	CSV flat file (predictions_log.csv)	PostgreSQL or IBM Db2 cloud database
Model	XGBoost (batch retrain)	Online learning with new incoming data
Frontend	Flask-rendered Jinja2 templates	React/Vue SPA with Flask REST API backend

Future Enhancement Roadmap

S.No	Enhancement	Description	Priority	Assigned To
1	Real-time API	Connect to IMD / OpenWeatherMap API for live data	High	Venkatesh Balireddy
2	SMS Alerts	Auto-notify registered authorities when flood is detected	High	Archana Dhanani
3	Multi-district Pred.	Batch predict flood risk for multiple districts	Medium	Shivatmika Gandikota

S.No	Enhancement	Description	Priority	Assigned To
4	LSTM Model	Replace XGBoost with LSTM for sequential rainfall analysis	Medium	Gode S R Durgaprasad
5	Mobile Application	Native Android / iOS app using the Flask REST API backend	Medium	Bavisetty Gopi Krishna
6	Regional Language UI	Multi-language support for Telugu, Hindi	Low	All Members